Ozone US-all Metric Inventory and Top-2 Summary

April 21, 2026

1 Source and reading guide

This note summarizes the following workbook:

> D:/Github/spatial-skew-t/code/analysis/ozone/US-all/output/us-all/tables/comparison_top2.updated.xlsx

The goal is to keep the metric definitions, ranking logic, and top-2 model configurations in one manuscript-ready LaTeX file.

In the workbook, `rank_value` is always the quantity used to order the top-2 rows, whereas `score_value` is the raw metric itself. The detailed tables below collapse each `ranking_group` into one row with a winner and a runner-up. Because every top-2 entry in this workbook has `CMAQ=yes`, `CMAQ` is documented in the column glossary but omitted from the per-row configuration strings.

2 Metric inventory

3 High-level patterns

The global probabilistic-score block still leans toward Morris-baseline skew-t settings. In particular, CRPS is led by settings 35 and 11, and many Brier and Quantile-score winners at low-to-moderate quantile levels remain in the Morris lane.

Upper-tail scoring is more heterogeneous. AR2 wins Brier at $q = 0.95$ and $q = 0.995$, and it also takes Quantile-score wins at $q = 0.98$, 0.99, and 0.995. MRTS wins Brier at $q = 0.9$, Quantile score at $q = 0.8$ and 0.9, and the below-threshold Brier split at $q = 0.98$ and 0.99.

Classification metrics show a precision-recall trade-off. High-threshold `Other` skew-t settings, especially settings 18 and 14 with `thr=85`, dominate precision and specificity. By contrast, MRTS appears most clearly in F1/Recall at $q = 0.98$, whereas calibration diagnostics remain mostly Morris or AR2: AR2 wins coverage at 0.8 and 0.9, and Morris still dominates most PIT-based summaries.

Block	Metrics	Scenario dimension	Top-2 ranking rule
Probabilistic score	Brier, Quantile score, CRPS	Brier thresholds $q \in \{0, 0.1, \ldots, 0.995\}$, quantile-score levels $q \in \{0.1, \ldots, 0.995\}$, and one overall CRPS summary	`ranking_basis = rel_to_gaussian`; smaller `rank_value` is better.
Tail-aware split score	Brier split	$q \in \{0.9, 0.95, 0.98, 0.99, 0.995\}$ crossed with `above_threshold`, `all`, and `below_threshold`	`ranking_basis = rel_to_gaussian`; smaller `rank_value` is better.
Classification	Accuracy, Precision, Recall, Specificity, F1	Event quantiles $q \in \{0.9, 0.95, 0.98, 0.99, 0.995\}$ with probability cutoff 0.5	`ranking_basis = raw_score`; the workbook already sorts in the correct direction for each metric.
Calibration to target	Coverage, PIT mean, PIT variance	Coverage levels $0.8, 0.9, 0.95$ plus overall PIT summaries	`ranking_basis = abs_gap_to_target`; smaller absolute gap is better.
Raw uncertainty diagnostics	PIT KS, PIT uniformity MAE, PIT uniformity RMSE	Overall summaries	`ranking_basis = raw_score`; lower values are better.

Table 1: Metric blocks represented in `comparison_top2.updated.xlsx`.

Workbook column(s)	Interpretation	How it appears in this draft
metric_family, metric	Identify the metric block and the named metric.	Used as the sectioning logic and the first column of each detailed table.
ranking_group	Source-side identifier for a specific metric/scenario combination.	Suppressed in the detailed tables because metric + scenario is easier to read in the manuscript.
target_level, threshold_value, target_value	Define the nominal quantile/coverage level and the corresponding event or target.	Condensed into the manuscript-side *Scenario* column.
prediction_cutoff	Probability cutoff used by classification metrics.	Included in the *Scenario* column for classification rows.
band_type	Split used by the Brier decomposition: above_threshold, all, or below_threshold.	Included in the *Scenario* column for Brier-split rows.
ranking_basis	Explains whether the workbook ranks by relative score, raw score, or absolute target gap.	Explained in the inventory table and used to interpret rank_value.
rank_value	Quantity actually used to assign rank 1 and rank 2.	Reported directly for both winner and runner-up.
score_value, score_se	Raw metric value and its standard error.	Reported as *Score* $\pm$ *SE*.
setting	Reproducible model identifier in the ozone US-all workflow.	Encoded inside each winner/runner-up label as S<setting>.
method, knots, thresh, TS, ar2, mrts	Core model-configuration fields.	Printed inside the winner/runner-up labels.
model_lane	Broad model family: Gaussian reference, Morris baseline, AR2, MRTS, or another numeric variant.	Printed inside the winner/runner-up labels as Gaussian, Morris, AR2, MRTS, or Other.
CMAQ	Indicator of whether CMAQ covariates are included.	Documented here only; all top-2 rows in this workbook have CMAQ=yes.

Table 2: Important workbook columns for manuscript use.

4 Detailed top-2 tables

Each winner/runner-up cell is a compact label containing the setting id, the model lane, the sampling family, and the key tuning fields `K`, `thr`, `TS`, `AR2`, and `MRTS`. The detailed tables retain the workbook's top-2 ordering but replace the source-side `ranking_group` string with a more compact scenario label.

4.1 Probabilistic scores

For this block, `rank_value` is the relative score with respect to the Gaussian reference, so smaller values indicate stronger performance.

Table 3: Top-2 probabilistic-score summaries from `comparison_top2.updated.xlsx`.

Metric	Scenario	Winner (setting/lane/model)	Rank value	Score ± SE	Runner-up (setting/lane/model)	Rank value	Score ± SE
Brier	q=0	S2 / Morris / max-stable (K=1, thr=75, TS=no, AR2=no, MRTS=-)	0.5834	0.00008342 ± 0.00008342	S104 / AR2 / skew-t (K=5, thr=0, TS=yes, AR2=yes, MRTS=-)	0.8289	0.0001185 ± 0.00007445
Brier	q=0.1	S33 / Morris / skew-t (K=6, thr=0, TS=no, AR2=no, MRTS=-)	0.9539	0.03689 ± 0.0009925	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.956	0.03697 ± 0.0009705
Brier	q=0.2	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9566	0.06051 ± 0.001701	S35 / Morris / skew-t (K=8, thr=0, TS=no, AR2=no, MRTS=-)	0.9572	0.06055 ± 0.00148
Brier	q=0.3	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9629	0.07603 ± 0.001555	S15 / Morris / skew-t (K=15, thr=0, TS=no, AR2=no, MRTS=-)	0.9631	0.07604 ± 0.001287
Brier	q=0.4	S36 / Morris / skew-t (K=9, thr=0, TS=no, AR2=no, MRTS=-)	0.969	0.08652 ± 0.0006096	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9692	0.08653 ± 0.0007625
Brier	q=0.5	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9802	0.09271 ± 0.001121	S15 / Morris / skew-t (K=15, thr=0, TS=no, AR2=no, MRTS=-)	0.9807	0.09276 ± 0.001229
Brier	q=0.6	S34 / Morris / skew-t (K=7, thr=0, TS=no, AR2=no, MRTS=-)	0.988	0.09345 ± 0.0003049	S104 / AR2 / skew-t (K=5, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9884	0.09348 ± 0.0004491
Brier	q=0.7	S122 / AR2 / skew-t (K=15, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9849	0.08519 ± 0.0006925	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9859	0.08528 ± 0.0004477
Brier	q=0.8	S16 / Morris / skew-t (K=15, thr=50, TS=no, AR2=no, MRTS=-)	0.9868	0.0706 ± 0.001406	S60 / Morris / skew-t (K=7, thr=0, TS=yes, AR2=no, MRTS=-)	0.9897	0.0708 ± 0.001739
Brier	q=0.9	S204 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=10)	0.9916	0.04467 ± 0.00126	S51 / Morris / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=-)	0.9922	0.0447 ± 0.001301
Brier	q=0.95	S111 / AR2 / skew-t (K=7, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9856	0.02664 ± 0.001991	S120 / AR2 / skew-t (K=10, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9886	0.02672 ± 0.002042
Brier	q=0.98	S58 / Morris / skew-t (K=6, thr=50, TS=yes, AR2=no, MRTS=-)	0.9734	0.01257 ± 0.0007027	S55 / Morris / skew-t (K=5, thr=50, TS=yes, AR2=no, MRTS=-)	0.9737	0.01257 ± 0.0006884
Brier	q=0.99	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.9466	0.006876 ± 0.0006893	S8 / Morris / skew-t (K=5, thr=50, TS=no, AR2=no, MRTS=-)	0.95	0.0069 ± 0.0005906
Brier	q=0.995	S124 / AR2 / t (K=15, thr=75, TS=yes, AR2=yes, MRTS=-)	0.9534	0.003842 ± 0.0001045	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.9556	0.003851 ± 0.0001188
CRPS	overall	S35 / Morris / skew-t (K=8, thr=0, TS=no, AR2=no, MRTS=-)	0.977	4.053 ± 0.01506	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9771	4.053 ± 0.01085
Quantile score	q=0.1	S72 / Morris / skew-t (K=15, thr=0, TS=yes, AR2=no, MRTS=-)	0.9927	4.366 ± 0.01051	S122 / AR2 / skew-t (K=15, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9927	4.366 ± 0.01304
Quantile score	q=0.2	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9959	7.223 ± 0.01741	S122 / AR2 / skew-t (K=15, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9959	7.223 ± 0.02098
Quantile score	q=0.3	S33 / Morris / skew-t (K=6, thr=0, TS=no, AR2=no, MRTS=-)	0.9975	9.185 ± 0.009203	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9976	9.185 ± 0.01037
Quantile score	q=0.4	S33 / Morris / skew-t (K=6, thr=0, TS=no, AR2=no, MRTS=-)	0.9982	10.4 ± 0.01359	S37 / Other / skew-t (K=11, thr=0, TS=no, AR2=no, MRTS=-)	0.9982	10.4 ± 0.01603
Quantile score	q=0.5	S37 / Other / skew-t (K=11, thr=0, TS=no, AR2=no, MRTS=-)	0.9983	10.9 ± 0.03013	S60 / Morris / skew-t (K=7, thr=0, TS=yes, AR2=no, MRTS=-)	0.9984	10.9 ± 0.03219

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Metric	Scenario	Winner (setting/lane/model)	Rank value	Score ± SE	Runner-up (setting/lane/model)	Rank value	Score ± SE
Quantile score	q=0.6	S60 / Morris / skew-t (K=7, thr=0, TS=yes, AR2=no, MRTS=-)	0.9988	10.69 ± 0.03775	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	0.999	10.7 ± 0.03958
Quantile score	q=0.7	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9988	9.74 ± 0.02604	S7 / Morris / skew-t (K=5, thr=0, TS=no, AR2=no, MRTS=-)	0.9992	9.743 ± 0.026
Quantile score	q=0.8	S204 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=10)	0.9992	7.936 ± 0.02265	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9993	7.936 ± 0.0005303
Quantile score	q=0.9	S201 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=5)	0.9983	5.033 ± 0.0188	S7 / Morris / skew-t (K=5, thr=0, TS=no, AR2=no, MRTS=-)	0.9984	5.033 ± 0.0007798
Quantile score	q=0.95	S61 / Morris / skew-t (K=7, thr=50, TS=yes, AR2=no, MRTS=-)	0.9957	2.988 ± 0.009827	S119 / AR2 / skew-t (K=10, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9957	2.988 ± 0.004428
Quantile score	q=0.98	S119 / AR2 / skew-t (K=10, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9861	1.423 ± 0.01749	S36 / Morris / skew-t (K=9, thr=0, TS=no, AR2=no, MRTS=-)	0.9863	1.424 ± 0.02184
Quantile score	q=0.99	S108 / AR2 / skew-t (K=6, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9718	0.797 ± 0.01481	S67 / Morris / skew-t (K=9, thr=50, TS=yes, AR2=no, MRTS=-)	0.9732	0.7981 ± 0.01733
Quantile score	q=0.995	S108 / AR2 / skew-t (K=6, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9412	0.4413 ± 0.01264	S114 / AR2 / skew-t (K=8, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9428	0.442 ± 0.01541

4.2 Tail-aware Brier split

This table separates Brier performance into `above_threshold`, `all`, and `below_threshold` bands. As above, smaller `rank_value` indicates better relative performance to the Gaussian reference.

Table 4: Top-2 Brier-split summaries from `comparison_top2.updated.xlsx`.

Metric	Scenario	Winner (setting/lane/model)	Rank value	Score ± SE	Runner-up (setting/lane/model)	Rank value	Score ± SE
Brier split	q=0.9, band=above	S2 / Morris / max-stable (K=1, thr=75, TS=no, AR2=no, MRTS=-)	0	0 ± 0	S4 / Morris / skew-t (K=1, thr=50, TS=no, AR2=no, MRTS=-)	0.9543	0.2766 ± 0.001711
Brier split	q=0.9, band=all	S204 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=10)	0.9916	0.04467 ± 0.00126	S51 / Morris / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=-)	0.9922	0.0447 ± 0.001301
Brier split	q=0.9, band=below	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.245	0.004428 ± 0.002788	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.44	0.007952 ± 0.004024
Brier split	q=0.95, band=above	S16 / Morris / skew-t (K=15, thr=50, TS=no, AR2=no, MRTS=-)	0.9317	0.3543 ± 0.01198	S38 / Morris / skew-t (K=6, thr=50, TS=no, AR2=no, MRTS=-)	0.9339	0.3551 ± 0.01365
Brier split	q=0.95, band=all	S111 / AR2 / skew-t (K=7, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9856	0.02664 ± 0.001991	S120 / AR2 / skew-t (K=10, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9886	0.02672 ± 0.002042
Brier split	q=0.95, band=below	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.4761	0.004014 ± 0.001834	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.6233	0.005256 ± 0.001358
Brier split	q=0.98, band=above	S71 / Morris / t (K=10, thr=75, TS=yes, AR2=no, MRTS=-)	0.8541	0.4612 ± 0.007674	S74 / Morris / t (K=15, thr=75, TS=yes, AR2=no, MRTS=-)	0.8559	0.4621 ± 0.003531
Brier split	q=0.98, band=all	S58 / Morris / skew-t (K=6, thr=50, TS=yes, AR2=no, MRTS=-)	0.9734	0.01257 ± 0.0007027	S55 / Morris / skew-t (K=5, thr=50, TS=yes, AR2=no, MRTS=-)	0.9737	0.01257 ± 0.0006884
Brier split	q=0.98, band=below	S207 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=15)	0.9877	0.002187 ± 0.0002172	S204 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=10)	0.9971	0.002208 ± 0.0002081
Brier split	q=0.99, band=above	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.8009	0.5233 ± 0.04107	S71 / Morris / t (K=10, thr=75, TS=yes, AR2=no, MRTS=-)	0.8157	0.533 ± 0.03861
Brier split	q=0.99, band=all	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.9466	0.006876 ± 0.0006893	S8 / Morris / skew-t (K=5, thr=50, TS=no, AR2=no, MRTS=-)	0.95	0.0069 ± 0.0005906
Brier split	q=0.99, band=below	S207 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=15)	0.9911	0.0007457 ± 0.00008788	S1 / Gaussian / gaussian (K=1, thr=0, TS=no, AR2=no, MRTS=-)	1	0.0007523 ± 0.00009287
Brier split	q=0.995, band=above	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.736	0.5599 ± 0.06917	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.764	0.5812 ± 0.06674
Brier split	q=0.995, band=all	S124 / AR2 / t (K=15, thr=75, TS=yes, AR2=yes, MRTS=-)	0.9534	0.003842 ± 0.0001045	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.9556	0.003851 ± 0.0001188
Brier split	q=0.995, band=below	S1 / Gaussian / gaussian (K=1, thr=0, TS=no, AR2=no, MRTS=-)	1	0.0002316 ± 0.000001534	S207 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=15)	1.01	0.0002338 ± 0.000002013

4.3 Classification metrics

For this block, `rank_value` equals the raw classification metric after the workbook has applied the correct ranking direction.

Table 5: Top-2 classification summaries from `comparison_top2.updated.xlsx`.

Metric	Scenario	Winner (setting/lane/model)	Rank value	Score ± SE	Runner-up (setting/lane/model)	Rank value	Score ± SE
Accuracy	q=0.9, cutoff=0.5	S116 / AR2 / skew-t (K=9, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9383	0.9383 ± 0.001983	S33 / Morris / skew-t (K=6, thr=0, TS=no, AR2=no, MRTS=-)	0.9383	0.9383 ± 0.002275
Accuracy	q=0.95, cutoff=0.5	S55 / Morris / skew-t (K=5, thr=50, TS=yes, AR2=no, MRTS=-)	0.9646	0.9646 ± 0.00339	S35 / Morris / skew-t (K=8, thr=0, TS=no, AR2=no, MRTS=-)	0.9645	0.9645 ± 0.003224
Accuracy	q=0.98, cutoff=0.5	S3 / Morris / skew-t (K=1, thr=0, TS=no, AR2=no, MRTS=-)	0.984	0.984 ± 0.00134	S120 / AR2 / skew-t (K=10, thr=50, TS=yes, AR2=yes, MRTS=-)	0.984	0.984 ± 0.001424
Accuracy	q=0.99, cutoff=0.5	S115 / AR2 / t (K=8, thr=75, TS=yes, AR2=yes, MRTS=-)	0.9918	0.9918 ± 0.0004617	S74 / Morris / t (K=15, thr=75, TS=yes, AR2=no, MRTS=-)	0.9917	0.9917 ± 0.0008787
Accuracy	q=0.995, cutoff=0.5	S1 / Gaussian / gaussian (K=1, thr=0, TS=no, AR2=no, MRTS=-)	0.9956	0.9956 ± 0.000001658	S15 / Morris / skew-t (K=15, thr=0, TS=no, AR2=no, MRTS=-)	0.9956	0.9956 ± 0.00008504
F1	q=0.9, cutoff=0.5	S33 / Morris / skew-t (K=6, thr=0, TS=no, AR2=no, MRTS=-)	0.645	0.645 ± 0.005291	S61 / Morris / skew-t (K=7, thr=50, TS=yes, AR2=no, MRTS=-)	0.6442	0.6442 ± 0.002312
F1	q=0.95, cutoff=0.5	S4 / Morris / skew-t (K=1, thr=50, TS=no, AR2=no, MRTS=-)	0.5744	0.5744 ± 0.01409	S55 / Morris / skew-t (K=5, thr=50, TS=yes, AR2=no, MRTS=-)	0.5728	0.5728 ± 0.02383
F1	q=0.98, cutoff=0.5	S203 / MRTS / t (K=1, thr=75, TS=yes, AR2=no, MRTS=5)	0.4652	0.4652 ± 0.004034	S65 / Morris / t (K=8, thr=75, TS=yes, AR2=no, MRTS=-)	0.4638	0.4638 ± 0.01459
F1	q=0.99, cutoff=0.5	S74 / Morris / t (K=15, thr=75, TS=yes, AR2=no, MRTS=-)	0.4068	0.4068 ± 0.05383	S203 / MRTS / t (K=1, thr=75, TS=yes, AR2=no, MRTS=5)	0.3948	0.3948 ± 0.02491
F1	q=0.995, cutoff=0.5	S9 / Morris / t (K=5, thr=75, TS=no, AR2=no, MRTS=-)	0.3405	0.3405 ± 0.04048	S24 / Other / t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.3311	0.3311 ± 0.04538
Precision	q=0.9, cutoff=0.5	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.8409	0.8409 ± 0.03817	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.806	0.806 ± 0.06422
Precision	q=0.95, cutoff=0.5	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.8028	0.8028 ± 0.04336	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.7658	0.7658 ± 0.02477
Precision	q=0.98, cutoff=0.5	S3 / Morris / skew-t (K=1, thr=0, TS=no, AR2=no, MRTS=-)	0.7555	0.7555 ± 0.05849	S113 / AR2 / skew-t (K=8, thr=0, TS=yes, AR2=yes, MRTS=-)	0.7446	0.7446 ± 0.05306
Precision	q=0.99, cutoff=0.5	S47 / Other / skew-t (K=6, thr=85, TS=no, AR2=no, MRTS=-)	0.7607	0.7607 ± 0.01071	S115 / AR2 / t (K=8, thr=75, TS=yes, AR2=yes, MRTS=-)	0.75	0.75 ± 0.03571
Precision	q=0.995, cutoff=0.5	S15 / Morris / skew-t (K=15, thr=0, TS=no, AR2=no, MRTS=-)	0.9583	0.9583 ± 0.04167	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9545	0.9545 ± 0.04545
Recall	q=0.9, cutoff=0.5	S2 / Morris / max-stable (K=1, thr=75, TS=no, AR2=no, MRTS=-)	1	1 ± 0	S61 / Morris / skew-t (K=7, thr=50, TS=yes, AR2=no, MRTS=-)	0.5677	0.5677 ± 0.004929
Recall	q=0.95, cutoff=0.5	S4 / Morris / skew-t (K=1, thr=50, TS=no, AR2=no, MRTS=-)	0.4814	0.4814 ± 0.01243	S25 / Other / t (K=15, thr=75, TS=no, AR2=no, MRTS=-)	0.4765	0.4765 ± 0.01385
Recall	q=0.98, cutoff=0.5	S203 / MRTS / t (K=1, thr=75, TS=yes, AR2=no, MRTS=5)	0.3541	0.3541 ± 0.006249	S65 / Morris / t (K=8, thr=75, TS=yes, AR2=no, MRTS=-)	0.3505	0.3505 ± 0.01039
Recall	q=0.99, cutoff=0.5	S74 / Morris / t (K=15, thr=75, TS=yes, AR2=no, MRTS=-)	0.2853	0.2853 ± 0.03944	S203 / MRTS / t (K=1, thr=75, TS=yes, AR2=no, MRTS=5)	0.2764	0.2764 ± 0.01415
Recall	q=0.995, cutoff=0.5	S9 / Morris / t (K=5, thr=75, TS=no, AR2=no, MRTS=-)	0.2325	0.2325 ± 0.02558	S26 / Other / t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.2325	0.2325 ± 0.02558

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Metric	Scenario	Winner (setting/lane/model)	Rank value	Score ± SE	Runner-up (setting/lane/model)	Rank value	Score ± SE
Specificity	q=0.9, cutoff=0.5	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.9962	0.9962 ± 0.002791	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.9922	0.9922 ± 0.005306
Specificity	q=0.95, cutoff=0.5	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.9973	0.9973 ± 0.001807	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.9955	0.9955 ± 0.001813
Specificity	q=0.98, cutoff=0.5	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.9981	0.9981 ± 0.0008954	S113 / AR2 / skew-t (K=8, thr=0, TS=yes, AR2=yes, MRTS=-)	0.998	0.998 ± 0.0005126
Specificity	q=0.99, cutoff=0.5	S57 / Morris / skew-t (K=6, thr=0, TS=yes, AR2=no, MRTS=-)	0.9995	0.9995 ± 0.00004179	S110 / AR2 / skew-t (K=7, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9994	0.9994 ± 0.0001681
Specificity	q=0.995, cutoff=0.5	S15 / Morris / skew-t (K=15, thr=0, TS=no, AR2=no, MRTS=-)	1	1 ± 0.00004189	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	1	1 ± 0.00004189

4.4 Uncertainty and calibration

Coverage, PIT mean, and PIT variance are ranked by absolute gap to target; PIT KS, PIT MAE, and PIT RMSE are ranked directly on the raw score scale.

Table 6: Top-2 uncertainty and calibration summaries from `comparison_top2.updated.xlsx`.

Metric	Scenario	Winner (setting/lane/model)	Rank value	Score ± SE	Runner-up (setting/lane/model)	Rank value	Score ± SE
Coverage	level=0.8	S117 / AR2 / skew-t (K=9, thr=50, TS=yes, AR2=yes, MRTS=-)	0.00008193	0.8001 ± 0.003869	S209 / MRTS / t (K=1, thr=75, TS=yes, AR2=no, MRTS=15)	0.001552	0.8016 ± 0.02369
Coverage	level=0.9	S113 / AR2 / skew-t (K=8, thr=0, TS=yes, AR2=yes, MRTS=-)	0.0002728	0.9003 ± 0.004716	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.0003933	0.8996 ± 0.007384
Coverage	level=0.95	S27 / Other / skew-t (K=2, thr=0, TS=no, AR2=no, MRTS=-)	0.00007324	0.9499 ± 0.0007276	S31 / Other / skew-t (K=4, thr=0, TS=no, AR2=no, MRTS=-)	0.0001149	0.9499 ± 0.0006859